

TUTORIAL -1

Aim: - Understand SDLC and STLC Phases for a Real-World Software Project and Identify Testing Activities Performed During Each Development Stage

Description:

To study and understand the phases of Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC) using a real-world software project and identify testing activities performed during each development stage.

Software development follows a structured approach known as the Software Development Life Cycle (SDLC), which includes Requirement Analysis, Design, Development, Testing, Deployment, and Maintenance. To ensure software quality, organizations perform various testing activities through the Software Testing Life Cycle (STLC), which includes Requirement Analysis, Test Planning, Test Case Development, Test Execution, Defect Reporting, and Test Closure.

In this practical, an Online Food Delivery Application Login Module is considered as a real-world example. Students will understand how testing activities are performed throughout the software development process and how SDLC and STLC work together to deliver quality software.

Problem Statement :

Develop a simple Login Authentication Module for an Online Food Delivery Application and study how SDLC and STLC phases are applied during software development. Identify testing activities performed during each stage and verify the functionality of the login system using valid and invalid credentials.

Algorithm: / Procedure :

Step 1: Start the program.

Step 2: Store valid username and password.

Step 3: Accept username from the user.

Step 4: Accept password from the user.

Step 5: Compare entered username and password with stored credentials.

Step 6: If both credentials match, display "Login Successful".

Step 7: Otherwise, display "Invalid Username or Password".

Step 8: Stop the program.

Input Code :

```
# -----  
# Practical No. 1  
# Title:  
# Understand SDLC and STLC Phases for a Real-World Software  
# Project and Identify Testing Activities Performed During  
# Each Development Stage  
#  
# Real-Time Example:  
# Online Food Delivery Application Login Module  
#  
# This program demonstrates a Login Authentication System  
# which can be tested during various SDLC and STLC phases.  
# -----  
  
# Store valid username in the system database  
valid_username = "admin"  
  
# Store valid password in the system database  
valid_password = "admin123"  
  
# Accept username from user  
username = input("Enter Username: ")  
  
# Accept password from user  
password = input("Enter Password: ")  
  
# Check whether entered credentials are correct  
if username == valid_username and password == valid_password:  
  
    # Display success message when credentials match  
    print("Login Successful")  
  
else:  
  
    # Display error message when credentials are invalid  
    print("Invalid Username or Password")  
  
# End of Program
```

Testing Activities Performed

SDLC Phase	Testing Activity
Requirement Analysis	Requirement Review
Design	Test Planning
Development	Unit Testing
Testing	Functional Testing
Deployment	Smoke Testing
Maintenance	Regression Testing

Observation:

1. SDLC provides a framework for software development.
2. STLC provides a framework for software testing.
3. Testing starts from the requirement phase itself.
4. Early defect detection reduces development cost.
5. Proper testing improves software quality and reliability.
6. Development and testing activities run in parallel throughout the project lifecycle.

Sample Output 1

Enter Username: admin
Enter Password: admin123

Login Successful

Sample Output 2

Enter Username: admin
Enter Password: test123

Invalid Username or Password

The screenshot shows the Programiz Python Online Compiler interface. The editor contains a Python script for a login system. The output window shows the program's execution with user input and a successful login message.

```
main.py
1 #
2 # Real-Time Example:
3 # Online Food Delivery Application Login Module
4 #
5 # This program demonstrates a Login Authentication System
6 # which can be tested during various SDLC and STLC phases.
7 # -----
8 # Store valid username in the system database
9 valid_username = "admin"
10
11 # Store valid password in the system database
12 valid_password = "admin123"
13
14 # Accept username from user
15 username = input("Enter Username: ")
16
17 # Accept password from user
18 password = input("Enter Password: ")
19
20 # Check whether entered credentials are correct
21 if username == valid_username and password == valid_password:
22     # Display success message when credentials match
23     print("Login Successful")
24 else:
25     # Display error message when credentials are invalid
26     print("Invalid Username or Password")
27
28 # End of Program
```

Output

```
Enter Username: admin
Enter Password: admin123
Login Successful

=== Code Execution Successful ===
```

Program Outcome :

1. Understand the complete SDLC process.
2. Understand the complete STLC process.
3. Identify testing activities associated with each SDLC phase.
4. Analyze real-world software projects from a testing perspective.
5. Understand the importance of software quality assurance.
6. Learn software testing documentation practices.
7. Understand the relationship between development and testing teams.
8. Apply SDLC and STLC concepts in software projects.
9. Improve knowledge of software development and testing workflows.
10. Develop practical understanding of software quality management.